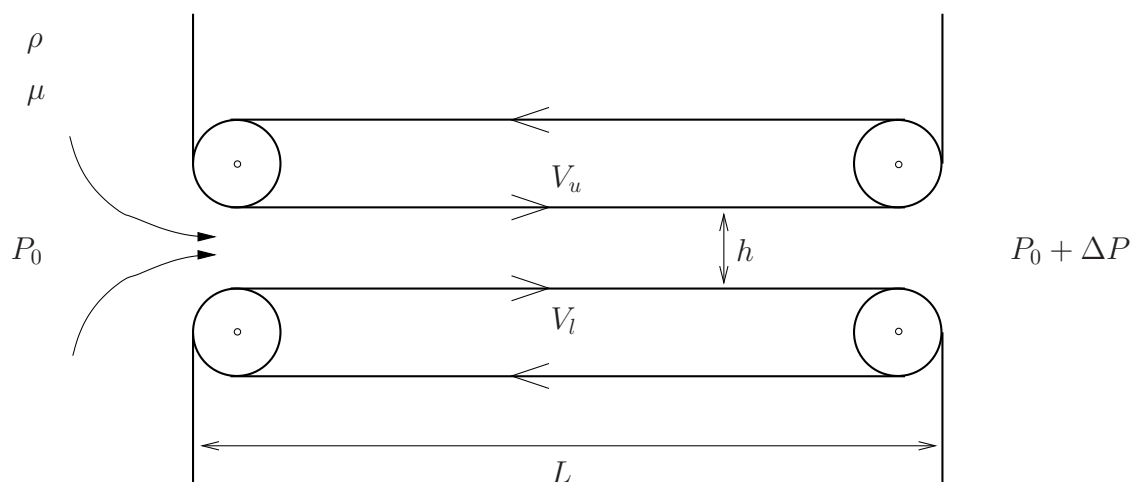


PROBLEM 4 (1 hour)

The system sketched in the figure is designed to steadily pump fluid of constant density ρ and constant viscosity μ from a reservoir, where the modified pressure has a constant value P_0 , to another one at pressure $P_0 + \Delta P$ ($\Delta P > 0$). To that end, a pair of parallel belts of length L , separated by a distance $h \ll L$, are driven at constant velocities V_l and V_u , dragging the liquid towards the large pressure reservoir.

The liquid flow can be assumed steady, two-dimensional (spanwise length much larger than h), and dominated by viscous forces in the gap. It proves convenient to introduce a cartesian coordinate system (x, y) , where x and y are the directions parallel and perpendicular to the belts, respectively. The **only** known parameters are $(\rho, \mu, P_0, \Delta P, L, h, V_l, V_u)$.



1. (1 point). Under the assumption of dominant viscous forces, obtain the characteristic velocity, $V_{\Delta P}$, induced in the liquid by the overpressure ΔP . Show that the pump will only work if the dimensionless parameter $\Lambda = (h^2 \Delta P) / [\mu L (V_l + V_u)]$, is sufficiently small.
2. (1 point). Provide the condition(s) that must be satisfied for the flow to be dominated by viscous forces.
3. (2 points). Determine the streamwise velocity profile in the gap between both belts, $v_x(x, y)$, as well as the volume flux per unit spanwise length, q . The results must be expressed as a function of known data, and of the reduced pressure gradient, P_l .
4. (2 points). Determine P_l , as well as the pressure distribution inside the channel, $P(x)$.
5. (2 points). Find the external forces per unit spanwise length in the x -direction, $F_{l,x}^{\text{EXT}}$ and $F_{u,x}^{\text{EXT}}$, that must be exerted on the lower and upper belts to keep the motion, as well as the power per unit spanwise length that needs to be supplied to drive the system, $\dot{W}$.
6. (2 points). Knowing that the net mechanical energy gained by the liquid per unit time and per unit spanwise length during the pumping process is $\dot{W}_{\text{useful}} = q \Delta P$, determine the efficiency of the pumping system, $\eta = \dot{W}_{\text{useful}} / \dot{W}$, as a function of the governing parameters. Study the particular cases a) $V_l = V_u$, and b) $V_l = 0$, demonstrating that in both cases the result can be expressed as a function of the single parameter Λ defined above. In case b), find the maximum efficiency, η_{max} as well as the corresponding value of Λ_{max} .
7. (1 additional point). Discuss the solution of Problem 3 in the light of the results obtained in Problem 4.

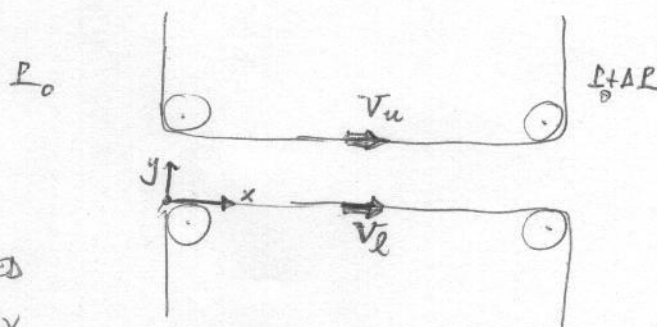
① CRITERION FOR DOMINANT VISCOUS FORCES

↳ TWO CHARACTERISTIC VELOCITIES IN THE PROBLEM $\begin{cases} V_e + V_u \\ V_{\Delta P} \text{ GIVEN BY THE BALANCE} \end{cases}$

$$\downarrow \frac{V_{\Delta P}}{V_e + V_u} = \frac{h^2 \Delta P}{\mu L (V_e + V_u)} = \Lambda$$

$$\frac{\Delta P}{L} \sim \mu \frac{V_{\Delta P}}{h^2} \Rightarrow V_{\Delta P} \sim \frac{h^2 \Delta P}{\mu L}$$

$$\begin{cases} \Lambda \ll 1 : \text{CRITERION: } \left[\frac{\rho (V_e + V_u) h^2}{\mu L} \ll 1 \right] \\ \Lambda \gg 1 : \left[\frac{\rho \Delta P h^4}{\mu^2 L^2} \ll 1 \right] \end{cases}$$



FOR THE FLUID TO ~~MOVE~~ BE PUMPED IT IS REQUIRED THAT Λ IS SMALL ENOUGH ($V_{\Delta P}$ SUFFICIENTLY SMALLER THAN $(V_u + V_e)$)

$$\textcircled{2} \quad \frac{\partial^2 v_x}{\partial y^2} = -\frac{P_e}{\mu} \Rightarrow v_x = -\frac{P_e y^2}{2\mu} + Ay + B$$

$y=0: v_x = V_e \Rightarrow B = V_e$
 $y=h: v_x = V_u \Rightarrow A = \frac{V_u - V_e}{h} + \frac{P_e h}{2\mu}$

THUS, $v_x = V_e + \frac{(V_u - V_e)}{h} y + \frac{P_e}{2\mu} y(h-y)$ THE FLOW RATE P.U. SPANWISE LENGTH IS

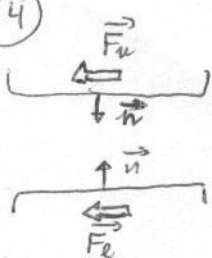
$$q = \int_0^h v_x(y) dy \Rightarrow q = \frac{(V_e + V_u)h}{2} + \frac{P_e h^3}{12\mu}$$

$$\textcircled{3} \quad \left[\frac{\partial q}{\partial x} = \int_0^h \frac{\partial v_x}{\partial x} dy = -\int_0^h \frac{\partial v_y}{\partial y} dy = v_y(0) - v_y(h) = 0 \right] \Rightarrow \frac{\partial P_e}{\partial x} = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} \frac{\partial^2 P}{\partial x^2} = 0 \Rightarrow P = P_0 + \Delta P \frac{x}{L} \\ P(x=0) = P_0 \\ P(x=L) = P_0 + \Delta P \end{cases} \Rightarrow \left[P_e = -\frac{\Delta P}{L} \right] \quad (P_e = -\frac{\partial P}{\partial x})$$

P.U. SPANWISE LENGTH

④ LET $\vec{F}_u$ AND $\vec{F}_l$ THE FORCES THAT THE LIQUID EXERTS ON THE UPPER AND LOWER BELTS, RESPECTIVELY. THEN



$$\begin{aligned} \vec{F}_u &= - \int_0^L p(-\vec{e}_y) dx + \int_0^L \vec{\tau}' \cdot (-\vec{e}_y) dx \rightarrow F_{ux} = - \int_0^L (\vec{e}_x \cdot \vec{\tau}' \cdot \vec{e}_y) dx \Big|_{y=h} \\ \vec{F}_l &= - \int_0^L p(\vec{e}_y) dx + \int_0^L \vec{\tau}' \cdot \vec{e}_y dx \rightarrow F_{lx} = \int_0^L (\vec{e}_x \cdot \vec{\tau}' \cdot \vec{e}_y) dx \Big|_{y=0} \end{aligned}$$

WE'LL ONLY CONSIDER THE FORCES IN THE X-DIRECTION

(2)

$$\vec{e}_x \cdot \vec{e}' \cdot \vec{e}_y = z'_{xy} = \mu \frac{\partial v_x}{\partial y} = \mu \frac{(V_u - V_l)}{h} - \frac{\Delta P}{2L} (h - 2y)$$

$$(\vec{e}_x \cdot \vec{e}' \cdot \vec{e}_y)|_{y=0} = \mu \frac{(V_u - V_l)}{h} - \frac{\Delta P h}{2L} \Rightarrow F_{lx} = \frac{\mu (V_u - V_l)L}{h} - \frac{\Delta P h}{2}$$

$$(\vec{e}_x \cdot \vec{e}' \cdot \vec{e}_y)|_{y=h} = \mu \frac{(V_u - V_l)}{h} + \frac{\Delta P h}{2L} \Rightarrow F_{ux} = -\frac{\mu (V_u - V_l)L}{h} - \frac{\Delta P h}{2}$$

FORCES THAT
THE FLUID EXERTS
ON THE BELTS

$$\left. \begin{aligned} F_{ux}^{(EXT)} &= -F_{ux} = \frac{\Delta P h}{2} + \frac{\mu (V_u - V_l)L}{h} \\ F_{lx}^{(EXT)} &= -F_{lx} = \frac{\Delta P h}{2} - \frac{\mu (V_u - V_l)L}{h} \end{aligned} \right\}$$

EXTERNAL
FORCES NEEDED TO DRIVE THE BELTS

POWER NEEDED TO MOVE THE BELTS P.U. SPANWISE LENGTH:

$$\dot{W} = F_{lx}^{(EXT)} V_l + F_{ux}^{(EXT)} V_u \Rightarrow \dot{W} = \frac{(V_l + V_u) h \Delta P}{2} + \frac{(V_u - V_l)^2 \mu L}{h}$$

$$\textcircled{5} \quad \eta = \frac{q \Delta P}{\dot{W}} \Rightarrow \eta = \frac{\frac{(V_l + V_u) h \Delta P}{2} - \frac{h^3 \Delta P^2}{12 \mu L}}{\frac{(V_l + V_u) h \Delta P}{2} + \frac{(V_u - V_l)^2 \mu L}{h}} \Rightarrow \eta = \frac{1 - \frac{\Lambda}{6}}{1 + \frac{(V_u - V_l)^2 \mu L}{(V_u + V_l)^2 \frac{2}{\Lambda}}}$$

a) $V_l = V_u$: $\eta = 1 - \frac{h^2 \Delta P}{6 \mu L (V_l + V_u)} \Rightarrow \eta = 1 - \frac{\Lambda}{6}$

b) $V_l = 0$: $\eta = \frac{1 - \frac{\Lambda}{6}}{1 + \frac{2}{\Lambda}}$

$\Lambda_{max}, \eta_{max}$ ACCOMPLISH THE CONDITION:

$$\left. \frac{d\eta}{d\Lambda} \right|_{\Lambda_{max}} = 0 \Rightarrow \Lambda_{max}^2 + 4\Lambda_{max} - 12 = 0 \Rightarrow \Lambda_{max} = 2$$

$$\eta_{max} = \eta(\Lambda_{max}) = \frac{1}{3}$$